

Connecting GitHub to VS Code

A Complete Step-by-Step Guide for Setup, Verification, and Daily Git Workflow

1 Prerequisites & Initial Sign-In

Before beginning, ensure **Git** is installed on your computer. Verify by running `git --version` in your terminal.

1. Open **Visual Studio Code**.
2. Click the **Accounts icon** (user silhouette) in the bottom-left corner of the window.
3. Select **Sign in with GitHub**.
4. Your web browser will open requesting authorization. Click **Authorize Visual Studio Code** and complete the login.
5. Return to VS Code. You are now authenticated with GitHub.

2 Configure & Verify Git Credentials

Set your global user identity so your commits are attributed correctly on GitHub:

1. Open Integrated Terminal: Press `Ctrl + ~` (or `Cmd + ~` on Mac).

2. Configure Identity: Run the following commands replacing with your details:

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

3. Verify Setup: Confirm your settings are saved correctly by executing:

```
git config --global user.name
git config --global user.email
```

Expected Result: The terminal displays your exact configured name and GitHub email address. Alternatively, run `git config --list` to inspect all settings.

3 Connecting a Repository

Choose one of the following methods depending on your starting point:

Option A: Clone an Existing GitHub Repository

1. Press `Ctrl` + `Shift` + `P` (or `Cmd` + `Shift` + `P`) to open the Command Palette.
2. Type `Git: Clone` and select it.
3. Select **Clone from GitHub** and pick your repository from the dropdown list.
4. Choose a local directory and click **Open**.

Option B: Publish a Local Folder to GitHub

1. Open your project folder in VS Code via `File > Open Folder...`.
2. Navigate to the **Source Control** tab (`Ctrl` + `Shift` + `G`).
3. Click **Publish to GitHub** and select **Public** or **Private**.

4 Daily Working Workflow (Saving & Syncing)

Local changes do **not** automatically update on GitHub. Follow this 3-step cycle in the **Source Control panel** whenever you make progress:

1. **Stage Changes** (`git add`): Hover over changed files in the *Changes* list and click the **+** (**Plus**) icon.
2. **Commit Changes** (`git commit`): Type a concise summary in the message field (e.g., `"Add user auth"`) and click **Commit** or press `Ctrl` + `Enter`.
3. **Push Changes** (`git push`): Click **Sync Changes** (or **Publish Branch**) to push your local commits online to GitHub.

Why Manual Syncing?

Keeping local work separate allows you to experiment, edit, and test code locally without publishing incomplete or broken code to your team's shared repository on GitHub.